

Gabriel Barrett

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SUMMARY

Computational Biologists with over 4 years of experience in analyzing biological data, developing bioinformatic pipelines, and applying machine learning techniques to genomic research. Skilled in multiple programming languages, with expertise in high-throughput sequencing data analysis, statistical modeling, and database management.

SKILLS

Data Organizing | Data Analytics | Data Mining | Data Integration | Data Visualization | Predictive Modeling | Statistics | GIS

Languages: Java, R, SQL, Bash, Git, NextFlow, Groovy, HTML, JavaScript, CS, Python

Tools and Frameworks: R Shiny, TRIPAL, Linux, GitHub, GitLab, Spring Boot, nf-core, SLURM, ArcGIS, NCBI/JGI/Ensembl databases, msprime, SLIM

EXPERIENCE

01/2024 – Present **Research Specialist**, Plant Computational Genomics – Storrs, CT

- Designed and implemented ETL pipelines in NextFlow to integrate genomic, transcriptomic, phenotypic, and environmental data into a PostgreSQL-based biological database based on the chado schema.
- Developed modular tools in Java and R for parsing and validating GFF3, VCF, and FASTA files for downstream integration into the DB schema.
- Used nf-core best practices to improve maintainability and conformance to community standards.
- Integrated workflows into a centralized analysis panel, enabling users to launch parameterized jobs on a HPC system for variant filtering, population structure, and genome-wide association via a web-based or CLI interface.

04/2024 – Present **Bioinformatic Consultant**, Eco-Evolution Lab – Storrs, CT

- Lead the analysis of a large-scale ddRAD-seq dataset (> 1.5TB, ~1500 samples) to investigate predate-prey dynamics across spotted salamander populations in a field experiment.
- Wrote a reproducible pipeline on nf-core GitHub for variant calling based on RADseq libraries.
- Advised on downstream data use for publication and grant reporting.

08/2021 – 01/2024 **Research Assistant**, Marine Evolutionary Ecology – Kingston, RI

- Executed molecular protocols for DNA extraction and library preparation, contributing to multiple genomics projects.
- Performed data analysis for collaborators interested in understanding sex determination mechanisms in the purple sea urchin and advised on data use for publication.
- Engineered robust scalable pipelines within NextFlow

EDUCATION

Graduated 05/2023 **Master of Science**, University of Rhode Island – Kingston, RI

Graduated 05/2021 **Bachelor of Arts**, University of Connecticut – Storrs, CT

PUBLICATIONS

Castellano, K. R., Neitzey, M. L., Starovoitov, A., **Barrett, G. A.**, Reid, N. M., Vuruputoor, V. S., ... & O'Neill, R. J. (2025). Genome Assembly of a Living Fossil, the Atlantic Horseshoe Crab *Limulus polyphemus*, Reveals Lineage-Specific Whole-Genome Duplications, Transposable Element-Based Centromeres, and a ZW Sex Chromosome System. *Molecular Biology and Evolution*, 42(2), msaf021.

Barrett, G. A. (2024). *Variant Graphs Improve Accuracy of Downstream Analysis in Restriction Site-Associated Sequencing*. University of Rhode Island.